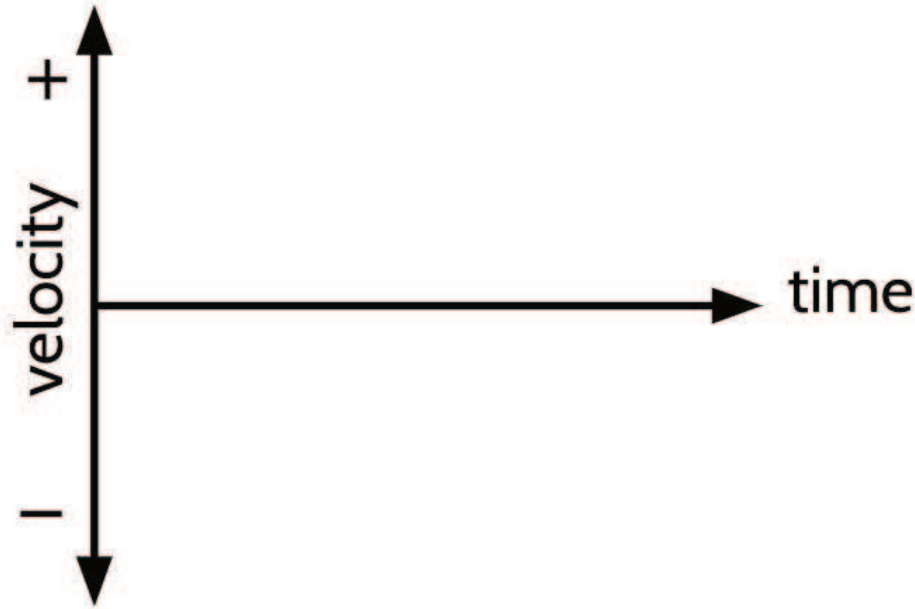


Name: _____

Date: _____

Momentum Change Investigation

Part 1: Use the graph below to show how the velocity of the cart would change from $t=0$ (when it starts traveling up the second ramp) to the time at which it rolls all the way back down it.



Part 2: Make a quantitative prediction: How would doubling the amount of force on the cart affect the time it takes to reach a velocity of zero going up the second ramp?

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Part 3: Collecting data. You can have more than one person collect data for each trial. If you do, record all these related values collected in the corresponding row. Conduct as many trials as you consider necessary. But first discuss with your group how many trials for each condition you plan to test in the 15 minutes available to collect data and which ramp heights you plan to test. Keep the following in mind:

- The magnitude of the force parallel to the cart along the ramp should approximately double when you change the end of ramp height from 3.5 inches to 7 inches.
- You either could use all of your trials to test only those two conditions or you could use up some of your trials to also test one or both intermediate conditions (4.5 inches and/or 5.5 inches).

Possible Ramp Height Conditions for this Investigation				
				
3.25" high brake ramp	4.75" high brake ramp	5.75" high brake ramp	6.5" high brake ramp	
Average net force acting on cart results from investigation				
Average Net Force (in N)				
Δt results from investigation				
Trial #	3.25" high brake ramp	4.75" high brake ramp	5.75" high brake ramp	6.5" high brake ramp

Part 4: Use CODAP to determine:

- What is the Δt for **each ramp height you tested** with your group?
 - 3.5 inches ramp: _____
 - 4.5 inches ramp: _____
 - 5.5 inches ramp: _____
 - 7.0 inches ramp: _____
- What is the Δt for **each ramp height you tested** as a class?
 - 3.5 inches ramp: _____
 - 4.5 inches ramp: _____
 - 5.5 inches ramp: _____
 - 7.0 inches ramp: _____

Part 5: Look back at your quantitative prediction that you made in Part 1. Do your results support or refute your prediction? Explain